Axcel®

AMLODIPINE-10mg TABLET

MAL11120002AZ

COMPOSITION:
Each tablet contains
Amlodipine Besylate
equivalent to Amlodipine 10mg

PRESENTATION:
White, rectangular tablet, 11.6mm in length and 8.3mm in width, with KOTRA engraved on one side and score - line on the opposite side.

INDICATIONS:
Amlodipine is indicated for the first line treatment of hypertension and can be used as the sole agent to control blood pressure in the majority of patients. Patients not adequately controlled on a single antihypertensive agent may benefit from the addition of amlodipine, which has been used in combination with the thiazide diuretic, alpha blockers, beta adrenoceptor blocking agent, or an angiotensin - converting enzyme inhibitor. Amlodipine is indicated for the first line treatment of myocardial ischemia, whether due to fixed obstruction (stable angina) and / or vasospasm / vasoconstriction (Prinzmetal's or variant angina) or coronary vasculature. Amlodipine may be used where the clinical presentation suggests a possible vasospastic / vasoconstrictive component but where vasospasm / vasoconstriction has not been confirmed. Amlodipine may be used alone, as monotherapy, or in combination with other antianginal drugs in patients with angina that is refractory to nitrates and / or adequate doses of beta blockers.

PHARMACOLOGY:
Mechanism of Action:
Amlodipine is a calcium ion influx inhibitor (slow channel-blocker or calcium ion antagonist) and inhibits the transmembrane influx of calcium ions into the cardiac and vascular smooth muscle. The mechanism of the antihypertensive action of amlodipine is due to a direct relaxant effect on the vascular smooth muscle. The precise mechanism by which amlodipine relieves angina has not been fully determined, but amlodipine reduces the total ischemic burden by the following two actions:
1) Amlodipine dilates the peripheral arterioles and thus, reduces the total peripheral resistance (afterload) against which the heart works. Since the heart rate remains stable, this unloading of the heart reduces myocardial energy consumption and oxygen requirements.

2) The mechanism of action of amlodipine probably involves dilation of the main coronary arteries and coronary arterioles, both in normal and ischemic regions. This dilatation increases myocardial oxygen delivery in patients with coronary artery spasm (Prinzmetal's or variant angina) and blunt smoking-induced coronary vasoconstriction.

In patients with hypertension, once-daily dosing provides clinically significant reductions of blood pressure in both the supine and standing positions throughout the 24-hr interval. Due to the slow onset of action, acute hypotension is not a feature of amlodipine administration. In patients with angina, once-daily administration of amlodipine increases total exercise time, time to angina onset, and time to 1 mm ST segment depression and decreases both angina attack frequency and nitroglycerin tablet consumption. Amlodipine has not been associated with any adverse metabolic effects or changes in plasma lipids and is suitable for use in patients with asthma, diabetes and gout.

Use in patients with Heart Failure:
Hemodynamic studies and a controlled clinical trial in NYHA Class II-IV heart failure patients have shown that amlodipine did not lead to clinical deterioration as measured by exercise tolerance, left ventricular ejection fraction and clinical symptomatology. A placebo controlled study (PRAISE) designed to evaluate patients in NYHA Class III-IV heart failure receiving digoxin, diuretics and ACE inhibitors has shown that amlodipine did not lead to an increase in risk of mortality or combined mortality and morbidity in patients with heart failure. In the same study, in a group of patients without clinical signs or symptoms suggestive of underlying ischemic disease, a clinically and statistically significant reduction in mortality and combined mortality and morbidity was observed with amlodipine.

Pharmacokinetics:
Absorption:
Amlodipine is well absorbed after oral doses with peak blood concentrations occurring after 6 to 12 hours. Absolute bioavailability has been estimated to be between 64 to 80%. Volume of distribution is approximately 21L/kg. Absorption of Amlodipine is unaffected by consumption of food.

Biotransformation / Elimination:
The terminal plasma elimination half-life is about 35 to 50 hours and is consistent with once daily dosing. Steady-state plasma level are reached after 7 to 8 days of consecutive dosing. Amlodipine is extensively metabolised by the liver to inactive metabolite with 10% of the parent compound and 60% of metabolites excreted in the urine.

Use in elderly:
The time to reach peak plasma concentrations of amlodipine is similar in elderly and younger subjects. Amlodipine clearance tends to be decreased with resulting increases in AUC and elimination half-life in elderly patients. Increases in AUC and elimination half-life in patients with congestive heart failure were as expected for the patient age group studied.

DOSAGE AND ADMINISTRATION:
For both hypertension and angina the usual initial dose is 5mg amlodipine once daily which may be increased to a maximum dose of 10mg depending on the individual patient's response. No dose adjustment of amlodipine is required upon concomitant administration of thiazide diuretics, beta-blockers, and angiotensin- converting enzyme inhibitors.

Use in the elderly:
Normal dosage regimens are recommended. Amlodipine, used at similar doses in elderly or younger patients, is equally well tolerated.

Use in Children:
Safety and effectiveness of amlodipine in children have not been established.

Use in Patients with Impaired Hepatic Function:
See section 4.4 Special Warning and Special Precautions for Use.

Use in Renal Failure:
Amlodipine may be used in such patients at normal doses. Changes in amlodipine plasma concentrations are not correlated with degree of renal impairment. Amlodipine is not dialyzable.

ROUTE OF ADMINISTRATION:
For Oral administration only.

CONTRAINDICATION:
Patient with known sensitivity to dihydropyridines, amlodipine or any of the inert ingredients.

PRECAUTION:
Used in patients with:
Impaired Hepatic Function: As with all calcium antagonists, amlodipine half-life is prolonged in patients with impaired liver function and dosage recommendations have not been established. The drug should therefore be administered with caution in these patients.

Effects on the Ability to Drive or Operate Machinery: Clinical experience with amlodipine indicates that it is unlikely to impair a patients ability to drive or use machinery.

Use in Pregnancy and Lactation:
Safety of amlodipine in human pregnancy and lactation has not been established. Amlodipine does not demonstrate toxicity in animal reproductive studies other than to delay parturition and prolonged labor in rats at a dose level 50 times the maximum recommended dose in human. Hence, use in pregnancy is only recommended when there is no safer alternative and when the disease itself carries greater risk for the mother and fetus.

INTERACTION WITH OTHER MEDICAMENTS:
Amlodipine has been safely administered with thiazide diuretics, alpha blockers, beta-blockers, angiotensin-converting enzyme inhibitors, long-acting nitrates, sublingual nitroglycerin, anti-inflammatory drugs, antibiotics and oral hypoglycemic drugs. *In vitro* data from studies with human plasma indicate that amlodipine has no effect on protein-binding of the drugs tested (digoxin, phenytoin, warfarin or indomethacin). In healthy male volunteers, the co-administration of amlodipine does not significantly alter the effect of warfarin on prothrombin response time. Pharmacokinetic studies with cyclosporin have demonstrated that amlodipine does not significantly alter the pharmacokinetics of cyclosporin.

Specials Studies: Effect of other agents on amlodipine

CIMETIDINE: Co-administration of amlodipine with cimetidine did not alter the pharmacokinetics of amlodipine.

ALUMINIUM / MAGNESIUM (Antacid): Co-administration of an aluminum/magnesium antacid with a single dose of amlodipine had no significant effect on the pharmacokinetics of amlodipine.

SILDENAFIL: A single 100mg dose of sildenafil in subjects with essential hypertension had no effect on the pharmacokinetic parameters of amlodipine. When amlodipine and sildenafil were used in combination, each agent independently exerted its own blood pressure lowering effect.

ATORVASTATIN: Co-administration of multiple 10mg doses of amlodipine with 80mg of atorvastatin resulted in no significant change in the steady state pharmacokinetic parameters of atorvastatin.

DIGOXIN: Co-administration of amlodipine with digoxin did not change serum digoxin levels or digoxin renal clearance in normal volunteers.

ETHANOL (alcohol): Single and multiple 10mg doses of amlodipine had no significant effect on the pharmacokinetics of ethanol.

WARFARIN: Co-administration of amlodipine with warfarin did not change the warfarin prothrombin response time.

CYCLOSPORIN: Pharmacokinetic studied with cyclosporin have demonstrated that amlodipine does not significantly alter the pharmacokinetics of cyclosporin.

SIDE EFFECTS:
Amlodipine is well tolerated. In patients with hypertension or angina, the most commonly observed side effects were headache, edema, fatigue, abdominal pain, nausea, flushing, somnolence, palpitation and dizziness. Less commonly observed side effects include dry mouth, increased sweating, back pain, malaise, weight increase / decrease, hypotension, syncope, hypertonia, hypoesthesia / paresthesia, peripheral neuropathy, tremor, gynecomastia. Others side effect like altered bowel habits, dyspepsia (including gastritis), gingival

hyperplasia, pancreatitis, vomiting, hyperglycemia, arthralgia, muscle cramps, myalgia, purpura, thrombocytopenia, impotence, insomnia, mood changes, coughing, dyspnea, rhinitis, alopecia, skin discoloration, urticaria, taste perversion, tinnitus, increased urinary frequency, micturition disorder, nocturia, vasculitis and visual disturbances. As with other calcium-channel blockers, the following adverse events have been rarely reported and cannot be distinguished from the natural history of the underlying disease, e.g. myocardial infarction, arrhythmia (including ventricular tachycardia and atrial fibrillation) and chest pain.

SYMPTOMS AND TREATMENT OF OVERDOSE:
In humans, experience with intentional overdose is limited. Available data suggest that gross overdosage could result in excessive peripheral vasodilation and possibly reflex tachycardia. Marked and probably prolonged systemic hypotension up to and including shock with fatal outcome have been reported. Administration of activated charcoal to healthy volunteer immediately after or up to two hours of amlodipine 10mg ingestion has been shown to significantly decrease amlodipine absorption. Gastric lavage may be worthwhile in some cases. Clinically significant hypotension due to amlodipine overdosage calls for active cardiovascular support including monitoring of cardiac and respiratory function, elevation of extremities, and attention to circulating fluid volume and urine output. A vasoconstrictor may be helpful in restoring vascular tone and blood pressure, provided that there is no contraindication to its use. Intravenous calcium gluconate may be beneficial in reversing the effects of calcium- channel blockade. Since amlodipine is highly protein-bound, dialysis is not likely to be of benefit.

STORAGE:
Store below 30^{O}C. Protect from light.

KEEP OUT OF REACH OF CHILDREN
JAUHI DARI KANAK-KANAK

PACK QUANTITIES:
Available in blister pack of 28’s, 30’s, 84’s, 100’s, 500’s and 1,000’s.
Not all pack sizes may be marketed.

Further information can be obtained from pharmacist, physician or the manufacturer.

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